Reliability

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Reliability Defined

- ***Reliability is "quality changing over time" or A motion picture instead of a snapshot***

Reliability Defined

- Reliability is a measure of the result of the quality of the product over the long run. Reliability terminates with a failure—i.e, unreliability occurs…
- Business enterprises observe the high cost of unreliability. The high cost of unreliability motivates an engineering solution to control and reduce costs.

Reliability Defined

- Reliability is an engineering discipline for applying scientific know-how to a component, assembly, plant, or process…
- so it will perform its intended function, without failure, for the required time duration when installed and operated correctly in a specified environment.

Reliability Defined

- **MIL-STD-721C Definitions of Terms for Reliability and Maintainability gives the following definition for reliability…**
- **Reliability is the probability than an item can perform its intended function without failure for a specified interval under stated conditions.**

Reliability Defined

This definition provides the following four aspects of reliability:

1. Reliability is a probability based concept; the numerical value of the reliability is between 0 and 1
2. The functional performance of the product has to meet certain stipulations and a functional definition of failure is needed. For example, a failure means different things to the user and to the repair person
3. It implies successful operation over a certain period of time…
4. Operating or environmental conditions under which product use takes place are specified

Designing for Reliability

Reliability does not just happen. It requires that the following three key elements be in place

- A commitment from top management to ensuring reliability
- A reliability policy (that goes hand-in-hand with a Quality Policy)…
- A philosophy that designs reliability in at an early stage

Ways to Improve Reliability

- Use Proven Design
- Use the simplest possible designs
- Use proven components that have undergone reliability component testing
  - Use redundant parts in high risk areas. Placing two components in parallel will reduce the overall probability of failure
  - Always design fail safe…
- Specify and use proven manufacturing methods

Measures of Reliability

Reliability is the probability that a system will still be functioning at time t.

$$R(t) = \frac{Number\ surviving\ at\ t}{Number\ existing\ at\ t = 0}$$

This can also be expressed as "the cumulative distribution of failure" …

$$F(t) = 1 - R(t) = \frac{Cummulative\ number\ of\ failure\ at\ t}{Number\ existing\ at\ t = 0}$$

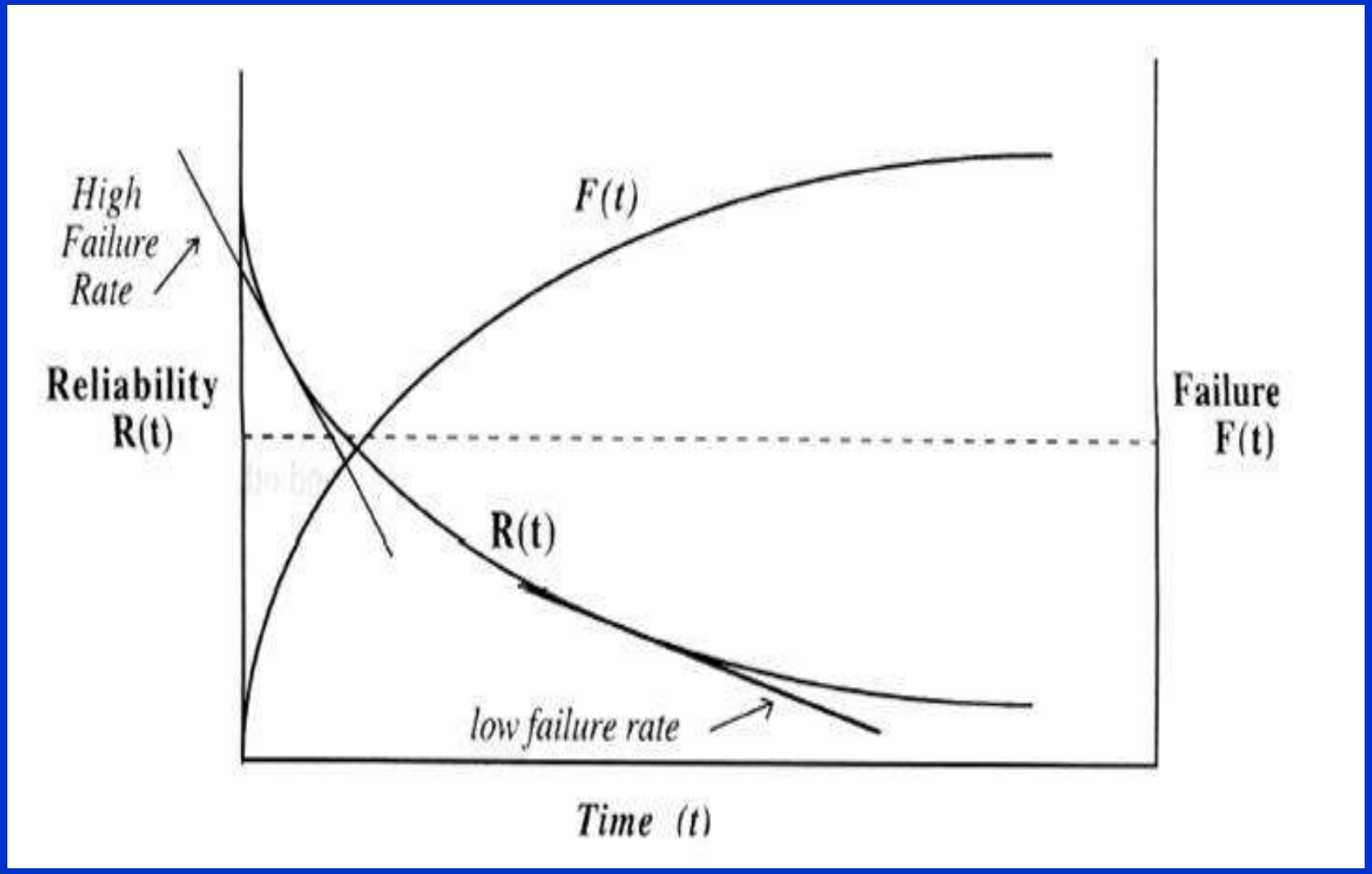
Measures of Reliability

These two measures are the mirror image of each other (Refer Figure below). The reliability will start at 1 and decay to approach 0 over time…

The cumulative distribution of failure will start at 0 (no failures) and approach 1 as all the items fail over time.

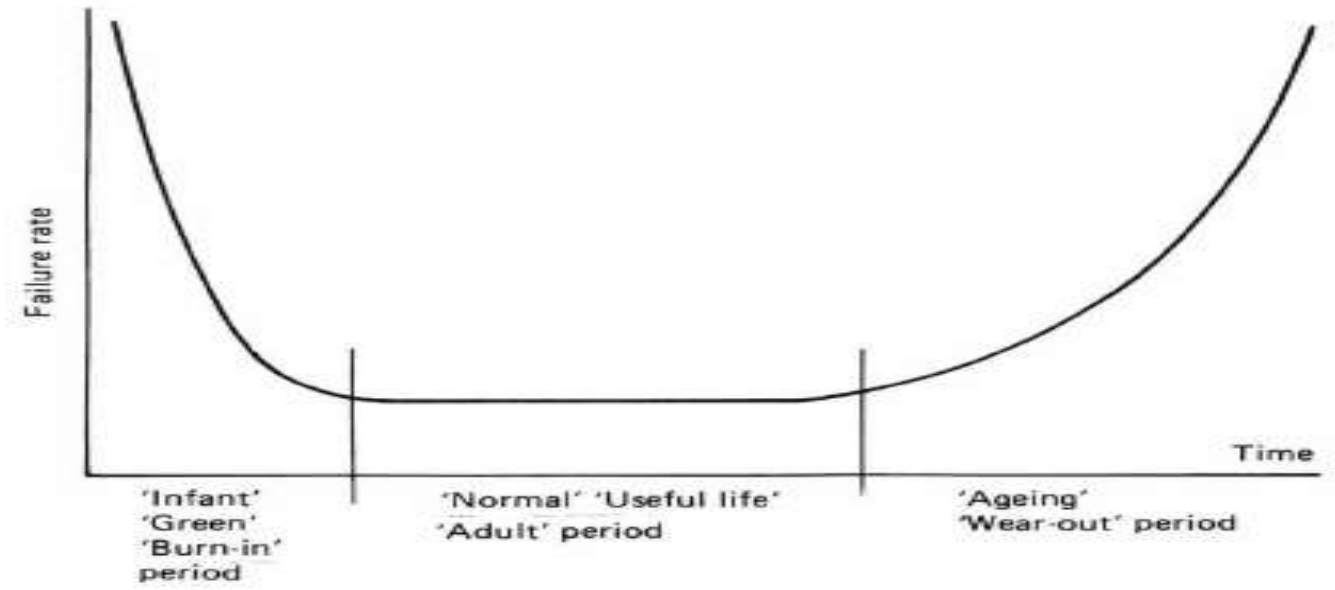
The slope of the reliability curve at any time t is the failure rate at that point in time. These measures give the overall reliability or failure at time t

Measures of Reliability



Bath Tub Curve

Most products go through three distinct phases from product inception to wear out. Figure below shows a typical life cycle curve for which the failure rate is plotted as function of time...



The bath-tub curve

Bath Tub Curve

Infancy/Green/Debugging/Burn-in-period:

Many components fail very soon after they are put into service. Failures within this period are caused by poor design that cause an item to be legitimately bad. These are called infant mortality failures and the failure rate in this period is relatively high. Good system vendors will perform an operation called "burn in" where they put together and test a system for several days to try to weed out these types of problems so the customer doesn't see them.

Bath Tub Curve

Chance failure / Normal Operating Life:

If a component does not fail within its infancy, it will generally tend to remain trouble-free over its operating lifetime. The failure rate during this period is typically quite low. This phase, in which the failure rate is constant, typically represents the useful life of the product.

Bath Tub Curve

Wear out /Ageing:

After a component reaches a certain age, it enters the period where it begins to wear out, and failures start to increase. The period where failures start to increase is called the wear out phase of component life.

Reliability Determination

In the adult or mature phase the failure rate of a component or system is constant. Under such conditions the time to failure follows an exponential law, and the probability density function of exponential distribution is as given below:

$$f(t) = \lambda e^{-\lambda t}$$

Where λ denotes the failure rate.

The Mean time to failure MTTF for the exponential distribution is…

$$MTTF = 1/\lambda$$

Reliability Determination

If the failure rate is constant, the MTTF is the reciprocal of the failure rate.

$$MTTF = 1/\lambda$$

For repairable system it is also equal to MTBF.

Reliability Determination

The reliability at time t, R(t), is the probability of the product lasting up to at least time t.

$$R(t) = 1 - F(t) = 1 - \int_0^t e^{-\lambda t} dt = e^{-\lambda t}$$

Reliability Determination-Example

Assume we have an automobile that is operating in its mature phase and has the following failure history:
Time to failure (hours): 100 800 1280 2600

The MTBF is given by: [100+800+1280+2600] / 4 = 1195 hours/failure

This gives a constant failure rate of: λ =1/1195 = 0.000836 failures/hour.

Reliability Determination-Example

What reliability can be expected from the automobile after 40, 200, 1000, and 5000 hours?

λ =1/1195 = 0.000836 failures/hour.

$$R(t) = 1 - F(t) = 1 - \int_0^t e^{-\lambda t} dt = e^{-\lambda t}$$

$$R(40) = e^{-0.000836x40} = 0.97 = 97\%$$

t-hours	40	200	1000	5000
Reliability	97%	84.6%	43.4%	1.5%

Reliability Determination-Example

Assume a minimum reliability of 0.9 can be accepted from the automobile, at what point would a service be required?

λ =1/1195 = 0.000836 failures/hour.

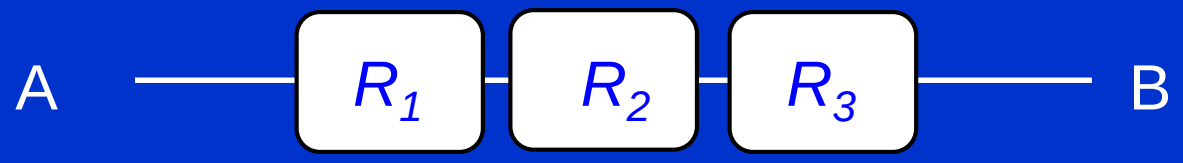
$$t = \frac{\ln R(t)}{-\lambda}$$

$$t = \frac{\ln(0.90)}{-0.000836} = 126\ hours$$

Thus, the automobile needs servicing every 126 hours to keep a minimum reliability of 0.9

System Reliability

Series System: When components are in series and each component has a reliability R_i. If one component fails, the system fails.



The overall reliability of a series system shown above is:

R_{AB} = R1 x R2 x R3

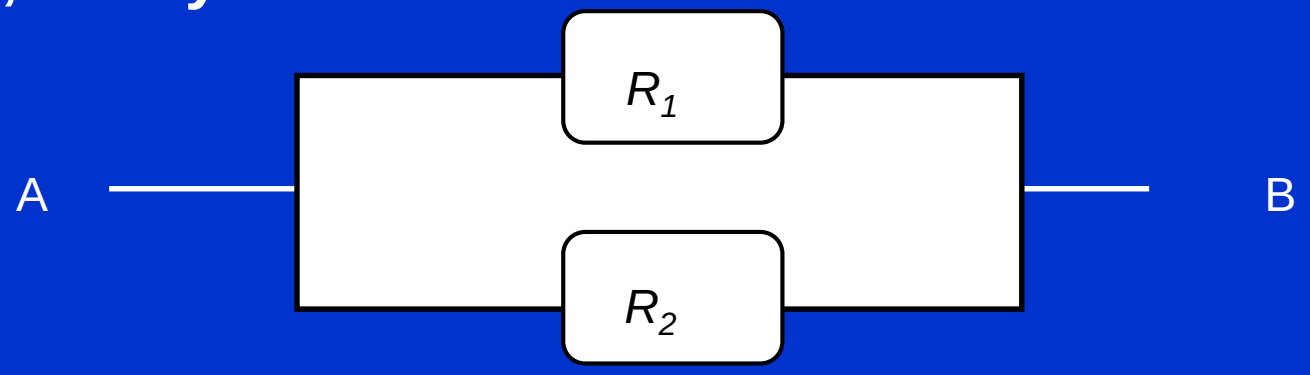
If $R_1 = R_2 = R_3 = 0.95$

$R_{AB} = R_1 \times R_2 \times R_3 = 0.95 \times 0.95 \times 0.95 = 0.86$

R_{total} is always < than R_1 or R_2 or R_3

System Reliability

Parallel System: When components are in parallel and each component has a reliability R_i. If one component fails, the system does not fail.



R_{AB} = 1 - probability (1 & 2 both fail)
The probability of 1 failing is = $(1 - R_1)$
The probability of 2 failing is = $(1 - R_2)$
Overall reliability is $R_{AB} = 1 - (1 - R_1)(1 - R_2)$
If $R_1 = 0.9$ and $R_2 = 0.8$
$R_{AB} = 1 - (1 - 0.9)(1 - 0.8) = 0.98$
R_{Total} is always > than R_1 or R_2

?

Thanks